Expectation Maximization (EM)

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References

- Wikipedia: `https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm`
- Stanford CS229 notes: `https://cs229.stanford.edu/main_notes.pdf`
- Caltech BE/Bi 103b: `https://bebi103b.github.io/lessons/optimization/em_algorithm.html`

1 Background

1.1 Theorems and Definitions

Theorem (Law of the unconscious statistician)**.** Let $p_X(x)$ be the probability mass function of a random variable X. Let $Y = g(X)$ be a random variable that is a function of X. Then, the expected value of Y is

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x)p_X(x)$$

Proof. (Adapted from Wikipedia) Assume that X takes on finitely many different values $x_1, x_2, ...$ The random variable $g(X)$ then has values $g(x_1), g(x_2), ...,$ although some of these may coincide with each other. Let $y_1, y_2, ...$ enumerate the possible distinct values of $g(X)$, and for each i let I_i denote the collection of all j with $g(x_j) = y_i$. Let $p_Y(y)$ denote the probability mass function of Y. Since a y_i can be the image of multiple distinct x_j,

$$p_Y(y_i) = \sum_{j \in I_i} p_X(x_j)$$

Then, according to the definition of expected value,

$$\begin{aligned}
\mathbb{E}[Y] = \mathbb{E}[g(X)] &= \sum_i y_i p_Y(y_i) \\
&= \sum_i y_i \sum_{j \in I_i} p_X(x_j) \\
&= \sum_i \sum_{j \in I_i} y_i p_X(x_j) \\
&= \sum_i \sum_{j \in I_i} g(x_j) p_X(x_j) \\
&= \sum_x g(x) p_X(x)
\end{aligned}$$

Following the conventions of CS 229 and Wikipedia, the notation used to explicitly indicate that the expected value of $Y = g(X)$ is expressed in terms of the probabiltiy distribution of X is $\mathbb{E}_{X \sim p_X}[g(X)]$. □

Definition (Evidence lower bound (ELBO))**.** Let X and Z be random variables jointly distributed with distribution $p(x, z \mid \theta)$. Then, for any distribution $q(z \mid x, \phi)$

$$\begin{aligned}
\mathbf{ELBO}(x, q, \theta) &= \sum_z q(z \mid x, \phi) \log \frac{p(x, z \mid \theta)}{q(z \mid x, \phi)} \\
&= \mathbb{E}_{z \sim q(z \mid x, \phi)} \left[\log \frac{p(x, z \mid \theta)}{q(z \mid x, \phi)} \right]
\end{aligned}$$

where the second equality relies on the law of the unconscious statistician.

Theorem (Jensen's inequality)**.** Let f be a convex function, and let X be a random variable. Then $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. Moreover, if f is strictly convex[1], then $\mathbb{E}[f(X)] = f(\mathbb{E}(X))$ if and only if X is a constant ($X = \mathbb{E}[X]$).

As a concrete example, if f is a scalar convex function, and X is a scalar random variable with probability mass function $p_X(x)$, Jensen's inequality states that

$$\sum_x (f(x) \cdot p_X(x)) \geq f\left(\sum_x (x \cdot p_X(x)) \right).$$

Applying the law of the unconscious statistician gives the shorthand notation $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$.

1.2 Problem setup

Given the statistical model which generates a set $\mathbf{X}$ of observed data, a set of unobserved latent data or missing values $\mathbf{Z}$, and unknown parameters θ (may be a vector), along with a likelihood function

$$L(\theta; \mathbf{X}, \mathbf{Z}) = p(\mathbf{X}, \mathbf{Z} \mid \theta)$$

[1] A strictly convex scalar function is defined by $f''(x) > 0$ for all x. A strictly convex vector-valued function has a positive definite Hessian, $H > 0$.

the maximum likelihood estimate (MLE) of the unknown parameters is

$$\arg\max_\theta L(\theta; \mathbf{X}) = \arg\max_\theta \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \theta) = \arg\max_\theta \sum_{\mathbf{Z}} p(\mathbf{X} \mid \mathbf{Z}, \theta) p(\mathbf{Z} \mid \theta)$$

2 CS 229

Consider the log-likelihood for a single sample x (and its corresponding latent variable z). Let Q be a **proposal distribution** over possible values of z.

$$\begin{aligned}
\log p(x \mid \theta) &= \log \sum_z p(x, z \mid \theta) \\
&= \log \sum_z Q(z) \frac{p(x, z \mid \theta)}{Q(z)} \\
&\geq \sum_z Q(z) \log \frac{p(x, z \mid \theta)}{Q(z)} \\
&= \mathrm{ELBO}(x, Q, \theta)
\end{aligned}$$

where the last step used Jensen's inequality. Specifically, $f(x) = \log x$ is a concave function ($f''(x) = -1/x^2 < 0$ over its domain $x \in \mathbb{R}_+$). Then,

$$f\left(\mathbb{E}_{z\sim Q}\left[\frac{p(x, z \mid \theta)}{Q(z)}\right]\right) \geq \mathbb{E}_{z\sim Q}\left[f\left(\frac{p(x, z \mid \theta)}{Q(z)}\right)\right]$$

where the $z \sim Q$ subscripts indicate that the expectations are with respect to z drawn from Q. To be more explicit about how Jensen's inequality is used here, let $g(Z)$ be the random variable

$$g(Z) = \frac{p(x, Z \mid \theta)}{Q(Z)}.$$

Note that Z is drawn according to Q. Then, Jensen's inequality is applied in the form

$$f(\mathbb{E}[g(Z)]) \geq \mathbb{E}[f(g(Z))]$$

This inequality ($\log p(x \mid \theta) \geq \mathrm{ELBO}(x, Q, \theta)$) holds for any distribution Q. We can make this inequality tight (i.e., hold with equality) if (as noted in Jensen's inequality theorem) $g(Z)$ is constant: let $Q(z) \propto p(x, z \mid \theta)$. Since Q is a distribution and $\sum_z Q(z) = 1$,

$$\begin{aligned}
Q(z) &= \frac{p(x, z \mid \theta)}{\sum_z p(x, z \mid \theta)} \\
&= \frac{p(x, z \mid \theta)}{p(x \mid \theta)} \\
&= p(z \mid x, \theta)
\end{aligned}$$

The EM algorithm alternatively updates Q and θ:

- E-step: Set Q to be the posterior distribution of z given x and the *current* parameters.

$$Q(z) = p(z \mid x, \theta^{(t)})$$

  To be explicit about the dependencies of Q, we can write $Q(z \mid x, \theta^{(t)})$ instead of $Q(z)$. This matches the form of $q(z \mid x, \phi)$ used in the definition above of the ELBO, with $\phi = \theta^{(t)}$.

- M-step: Maximize the ELBO function with respect to θ while *fixing* the choice of Q.

$$\begin{aligned}
\theta^{(t+1)} &= \arg\max_{\theta} \mathrm{ELBO}(x, Q, \theta) \\
&= \arg\max_{\theta} \sum_z Q(z \mid x, \theta^{(t)}) \log \frac{p(x, z \mid \theta)}{Q(z \mid x, \theta^{(t)})} \\
&= \arg\max_{\theta} \sum_z Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) \underbrace{- \sum_z Q(z \mid x, \theta^{(t)}) \log Q(z \mid x, \theta^{(t)})}_{\text{entropy of } Z\colon H(Z|x,\theta^{(t)})} \\
&= \arg\max_{\theta} \sum_z Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) \\
&= \arg\max_{\theta} \mathbb{E}_{z \sim Q} \log p(x, z \mid \theta)
\end{aligned}$$

  - Notice how the entropy of $Z \mid x, \theta^{(t)}$ does not depend on θ, so that term can be ignored. This makes the CS 229 expression for the M-step equivalent to the Wikipedia expression for the M-step.
  - The M-step can be interpreted as maximizing the expected value (over the proposal distribution) of the "complete-data" log-likelihood.

- CS 229 appears to follow an unconventional split of the E- and M-steps. In the conventional description of the EM algorithm (both in the orignal Dempster et al. (1977) paper and in Wikipedia), the E-step (*expectation step*) defines the expected value function

$$\mathbb{Q}(\theta \mid \theta^{(t)}) = \mathbb{E}_{z \sim Q} \log p(x, z \mid \theta) = \sum_z Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta)$$

  and the M-step (*maximization step*) finds the parameters θ that maximize that expected value function

$$\theta^{(t+1)} = \arg\max_{\theta} \mathbb{Q}(\theta \mid \theta^{(t)}).$$

Generalized to a dataset consisting of multiple samples $\{x^{(1)}, ..., x^{(n)}\}$, the log likelihood becomes

$$\begin{aligned}
\ell(\theta) &= \sum_{i=1}^{n} \log p(x^{(i)} \mid \theta) \\
&= \sum_{i=1}^{n} \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)} \mid \theta) \\
&\geq \sum_{i=1}^{n} \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)} \mid \theta)}{Q_i(z^{(i)})} \\
&= \sum_{i=1}^{n} \mathrm{ELBO}(x^{(i)}, Q_i, \theta)
\end{aligned}$$

Here, we introduce one distribution Q_i for each samples $x^{(i)}$ and set each Q_i to be the posterior distribution

$$Q_i(z^{(i)}) = p(z^{(i)} \mid x^{(i)}, \theta)$$

The full EM algorithm is therefore as follows:

1. Initialize $\theta^{(0)}$.
2. Repeat until convergence (either the ELBO stops improving, or the parameter values stop changing):
   - E-step: For each i, set
   $$Q_i(z^{(i)}) = p(z^{(i)} \mid x^{(i)}, \theta^{(t)})$$
   - M-step: Set
   $$\begin{aligned}
   \theta^{(t+1)} &= \arg\max_{\theta} \sum_{i=1}^{n} \mathrm{ELBO}(x^{(i)}, Q_i, \theta) \\
   &= \arg\max_{\theta} \sum_{i=1}^{n} \sum_{z^{(i)}} Q(z^{(i)} \mid x^{(i)}, \theta^{(t)}) \log \frac{p(x^{(i)}, z^{(i)} \mid \theta)}{Q(z^{(i)} \mid x^{(i)}, \theta^{(t)})} \\
   &= \arg\max_{\theta} \sum_{i=1}^{n} \sum_{z^{(i)}} Q(z^{(i)} \mid x^{(i)}, \theta^{(t)}) \log p(x^{(i)}, z^{(i)} \mid \theta) \\
   &= \arg\max_{\theta} \sum_{i=1}^{n} \mathbb{E}_{z^{(i)} \sim Q_i} \log p(x^{(i)}, z^{(i)} \mid \theta)
   \end{aligned}$$

3 BE/Bi 103b

While CS 229 describes EM primarily as a substitute for MLE when there are latent variables, it can be used to find other estimators, such as the maximum a posteriori (MAP) estimator. This is the approach taken in the BE/Bi 103b notes and is also briefly mentioned in the original Dempster et al. (1977) paper.

The marginal posterior distribution of interest is

$$p(\theta \mid x) = \frac{p(x, \theta)}{p(x)}$$

The MAP estimate is the value of θ that maximizes $p(\theta \mid x)$. Since the denominator $p(x)$ is not a

function of θ, the objective function is

$$\begin{aligned}
h(\theta) &= \log p(x, \theta) \\
&= \log \sum_z p(x, z, \theta) \\
&= \log \sum_z p(x, z \mid \theta) p(\theta) \\
&= \log \sum_z Q(z) \frac{p(x, z \mid \theta) p(\theta)}{Q(z)} \\
&\geq \sum_z Q(z) \log \frac{p(x, z \mid \theta) p(\theta)}{Q(z)}
\end{aligned}$$

where $p(\theta)$ is the prior distribution of the parameter values. To make this inequality tight, set $Q(z) = p(z \mid x, \theta)$ such that

$$\begin{aligned}
\sum_z Q(z) \log \frac{p(x, z \mid \theta) p(\theta)}{Q(z)} &= \sum_z p(z \mid x, \theta) \log \frac{p(x, z \mid \theta) p(\theta)}{p(z \mid x, \theta)} \\
&= \sum_z p(z \mid x, \theta) \log p(x, \theta) \\
&= \log p(x, \theta) \sum_z p(z \mid x, \theta) \\
&= \log p(x, \theta)
\end{aligned}$$

where the second equality comes from the fact that $p(x, z, \theta)$ is equal to both $p(z \mid x, \theta) p(x, \theta)$ and $p(x, z \mid \theta) p(\theta)$, so we can express $p(x, \theta)$ as

$$p(x, \theta) = \frac{p(x, z \mid \theta) p(\theta)}{p(z \mid x, \theta)}.$$

Thus, the E- and M-steps are

- E-step: Set $Q(z \mid x, \theta^{(t)}) = p(z \mid x, \theta^{(t)})$.

- M-step: Maximize the surrogate function

$$
\begin{aligned}
\theta^{(t+1)} &= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \log \frac{p(x, z \mid \theta) p(\theta)}{Q(z \mid x, \theta^{(t)})} \\
&= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) p(\theta) \underbrace{- \sum_{z} Q(z \mid x, \theta^{(t)}) \log Q(z \mid x, \theta^{(t)})}_{\text{entropy of } Z\colon H(Z|x,\theta^{(t)})} \\
&= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) p(\theta) \\
&= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \left(\log p(x, z \mid \theta) + \log p(\theta)\right) \\
&= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) + \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(\theta) \\
&= \arg\max_{\theta} \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) + \log p(\theta) \sum_{z} Q(z \mid x, \theta^{(t)}) \\
&= \arg\max_{\theta} \log p(\theta) + \sum_{z} Q(z \mid x, \theta^{(t)}) \log p(x, z \mid \theta) + \\
&= \arg\max_{\theta} \log p(\theta) + \mathbb{E}_{z \sim Q} \log p(x, z \mid \theta)
\end{aligned}
$$